

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)		Kills bacteria	1			1		
		(ii)		• (Gram positive) {have (thick) peptidoglycan in cell wall / no additional lipopolysaccharides} (so more susceptible to antibiotic) (1) • Inhibits formation of cross linkages between {peptidoglycans/ amino acids} (1) • (peptidoglycan) cell wall is {broken down/ digested} /Cell wall is weakened (1) • {Osmosis/ water moves in} causing cell to {lyse/ burst} (1)	1	2	1	4		1
		(iii)		Any three (×1) from: • Reference to mutation (1) • (means some bacteria) become <u>resistant</u> (1) • (The resistant bacteria) have a selective advantage /are able to survive antibiotic (1) • Resistance passed on (when bacteria divide/ reproduce/to next generation) (1)		3		3		
		(iv)		Don't scratch bites/ wash area/ use of anti-bacterial cream/ AVP Reject antibiotics/ insect repellents			1	1		
	(b)	(i)		Endemic – (always) present at low levels, epidemic – significant increase (in usual number of cases) (1) Pandemic – affects a large number of people/occurs {over a wide area/ many countries/ global} (1)	1		1	2		

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		(ii)		Any two (x1) from: Different species (of <i>Plasmodium</i>) (1) Different antigen(ic types) (1) Mutates regularly (1)	2			2		
	(c)	(i)		less vaccination			1	1		1
		(ii)		73500 = 2 marks If incorrect award 1 mark for 10.5/100 x 700000 Allow 1 mark for 63000 (figure for 2018 instead of 2020)		2		2	2	
		(iii)		Any three (x1) from: - Memory cells already present (1) - Due to prior exposure to antigen (1) - Memory cells detect antigen (1) - Shorter latent period (1) - Memory cells divide faster if same antigens are encountered (1)	1	2		3		
		(iv)		1994/ 1995/ 1996 Accept 2016			1	1		1
				Question 7 total	6	9	5	20	2	3